

Executive Summary

Probabilistic Risk Assessment: NASA and ESA use structured PRA methods (event trees, fault trees, Monte Carlo) alongside traditional hazard analysis ¹ ². NASA's PRA standard (NPR 8705.5A) prescribes quantifying subsystem failure rates and system-level loss-of-mission probabilities (targets ~0.1%). ESA historically uses deterministic hazard analysis (FMEA/FTA) with probabilistic estimates of residual risk ². Launch vehicles exhibit very high reliability today: for example Atlas V's record is 100% mission success in 100 flights ³ (~99.96% with confidence); SpaceX's Falcon 9 family ~99.54% ⁴; Ariane 5 ~96% ⁵ (117 flights). (Ariane 6 is new, with limited data.) Published analyses of deep-space probes show roughly half of Mars missions have failed historically ⁶, often in cruise (e.g. communications/loss, propulsion failures) or at arrival (orbit insertion or landing failures). Outer-planet missions have fared better (Voyagers, Galileo, Cassini succeeded; Galileo's high-gain antenna failure was partial). Sample-return missions are mixed (Stardust and Hayabusa succeeded; Genesis reentry failed). Detailed per-phase failure statistics are scarce in open literature, but several reviews note ~50% overall failure on Mars ⁶ and higher reliability for larger NASA flagship missions. Subsystem lifetimes vary: e.g. avionics/computers often have MTTF on the order of 10^5 – 10^6 hours (tens of years), while fluid systems (propulsion valves, thrusters) may be more limited (10^3 – 10^4 hours) under stress. Standard references (e.g. Wertz's SMAD) and NASA reliability handbooks give orders-of-magnitude (e.g. relays, batteries ~ 10^4 – 10^5 hr, reaction wheels ~ 10^6 rev) but actual values depend on design. In decision analysis, mission "value" is typically risk-adjusted by multiplying payoff by success probability (or using risk-neutral probabilities in a decision tree) ⁷. For example, one study derives a risk-neutral weighting $q=(1-d)/(u-d)$ and computes expected NPV under uncertainty ⁷. In practice, analysts often compute an expected return $\approx p_{\text{success}} \times (\text{benefit}) - \text{cost}$, or run Monte Carlo on costs and revenues. Mission insurance beyond launch is virtually nonexistent for deep-space probes. A 2025 industry report notes that satellite insurance is very expensive and many operators self-insure ⁸. Only ~300 of ~12,787 active satellites are insured for in-orbit accidents ⁸. By analogy, science probes are normally "self-insured": e.g. Hayabusa, OSIRIS-REx, and Cassini bore all mission risk internally, aside from required launch liability insurance. European operators must legally insure satellites, but even so ESA missions typically carry risk in budgets rather than buy expensive insurance. **Redundancy strategies** (multiple subsystems or spacecraft) are used when high reliability is needed. Common approaches include cross-strapping (dual-string electronics), spare instruments, and entire second spacecraft or backup launches. For example, NASA's GRAIL, Van Allen probes, and the dual-sats in Galileo gravity (GRAIL) or twin-sat asteroid missions demonstrate parallel architectures. Reliability theory shows diminishing returns: adding a redundant unit doubles cost but only modestly raises success probability ($P=1-(1-p)^2$). Nonetheless, for some designs a dual-craft or dual-launch (rideshare) is considered to *vet* outcomes. Literature (e.g. engineering trade studies) analyzes cost-vs-reliability curves: generally, each $1\times$ mass/cost of redundancy buys a fractional increase in success probability, so mission planners balance mass/cost against acceptable risk.

Probabilistic Risk Assessment (PRA) Techniques

- **PRA Methodology:** NASA's PRA procedures (NPR 8705.5A) call for fault-tree/event-tree modeling and Monte Carlo simulation of mission sequences ¹. Systems are decomposed into critical events (e.g. loss of propulsion, comms failure), each with a probability or failure rate. Monte Carlo "risk

spectrum” runs random fault injections through the mission timeline. ESA’s approach (e.g. as of 1989) was to use hazard analysis to assign risks qualitatively and then apply PRA for residual risk ² . In both agencies, PRA complements deterministic safety analysis. Typical practice includes Failure Modes and Effects Analysis (FMEA) to identify weak points, and Fault Tree Analysis (FTA) to compute system failure probability. For deep-space missions, PRA often covers phases like launch, cruise, approach, entry/descent, and surface (if relevant), with separate event trees for each. Mission assurance teams at NASA/JPL routinely compute *Loss of Mission* probabilities; historically they target <0.001 (0.1%) for a mission-critical loss ⁹ . These PRA results inform design choices (redundancy levels) and risk mitigation (e.g. adding warm spares for thrusters, duplicating critical avionics).

- **Launch Vehicle Reliability:** Key launch vehicles have well-documented track records. Atlas V (100 flights to mid-2024) is at 100% mission success ³ . Falcon 9 (Block 5 era) has achieved ~99.54% ⁴ (649/652 successes) with a 95% confidence interval of roughly 98.9–99.96%. Ariane 5 (117 flights) has ~96% success ⁵ (about 5 failures) with 95% CI ~91.5–99%. Falcon Heavy has flown 12 times (Feb 2018–2026) with all successes (nominally 100%, though CI ~73–100%). (By contrast older rockets like Delta IV Heavy had 15/16 successes ~94%.) These statistics come from launch databases and publications. Agencies like NASA LSP maintain internal reliability databases, but open sources (manufacturer data, academic analyses) suffice. For portfolio planning, one uses the reliability *distribution* of launch: e.g. SpaceX publishes manifest but not formal reliability; we combine known failures with binomial confidence intervals. Key references: Wikipedia “List of Falcon launches” provides numbers (citable here ⁴); the French press notes Ariane successes ⁵ . An open-source mission planner should allow input of per-launch success probability (with uncertainty) and propagate it into mission success.

- **Mission-Phase Reliability:** Historical data can be mined for per-phase failure probabilities. For Mars missions, Planetary Society notes roughly half fail overall ⁶ . Qualitatively: of all Mars probes sent, failures occurred both en-route (e.g. *Mars Observer* lost on cruise, *Phobos-Grunt* in Earth orbit) and at arrival (e.g. *Mars Climate Orbiter* wrong unit, *Mars Polar Lander* descent). An open-sources analysis (Bright Ascension 2019) reports ~87% success (2009–2018) for small satellites ¹⁰ , suggesting recent reliability gains, but deep space has higher risks. Outer-planet missions have mixed outcomes: Pioneers 10/11 (1970s) achieved cruise flybys (contact lost later for 10), Voyagers 1/2 succeeded fully, Galileo succeeded despite an antenna glitch, Cassini fully succeeded its mission, Juno is ongoing (no failures so far). Sample-return history: *Genesis* (2001) failed in EDL (re-entry parachute); *Stardust* (2006) and *Hayabusa* (2010) succeeded (though Hayabusa had non-nominal returns via ion thrust); *OSIRIS-REx* (2023) succeeded. Technology demos (e.g. Deep Space 1, Hayabusa’s ion engine) have also been risky but tended to finish primary goals. **Literature:** Formal mission success reviews are published by NASA/ESA (e.g. “Overview of OSIRIS-REx Mission” in Space Sci. Rev.) but seldom quantify cruise vs arrival failure rates. A few PhD theses and conference papers analyze aggregate success (often only stating “<50% Mars success”). Our planner should include an adjustable failure-probability by mission phase (e.g. set “orbital insertion success = 0.95”) based on gathered data.

- **Subsystem Reliability (MTBF):** Published values are rare, but some data exist. NASA’s Reliability Standards (e.g. NASA-STD-8729.1) categorize components by quality grade (requirements of “no single point failure”) ⁹ . Spacecraft parts reliability is often predicted by MIL-HDBK-217 or its successors, giving e.g. electronics (μ C, FPGAs) MTBF $\sim 10^5$ – 10^6 h; rad-hard power converters $\sim 10^4$ h; mechanical relays $\sim 10^4$ h; reaction wheels $\sim 10^6$ rev; ion engine thruster lifetimes thousands of hours. SMAD (Space Mission Analysis & Design) cites generic values from historical

missions: communications amplifiers (traveling-wave tubes) $\sim 10^5$ h, star trackers $\sim 50,000$ h (before overhaul), batteries ~ 5 – 10 years, etc. JPL often maintains institutional databases (usually classified) but some surveys (e.g. “Spacecraft Bus Reliability” by Weier et al.) provide summary. For design, a rule-of-thumb table (like MIL-HDBK-217F) is used: e.g. μ processor MTTF $\sim 500,000$ h, FPGA $\sim 250,000$ h, board power converter $\sim 50,000$ h, thruster valve $\sim 10,000$ h. In absence of exact data, our planner should allow entering custom MTBFs or failure rates for each subsystem and propagate them via FTA.

- **Risk-Adjusted Value (rNPV):** The planning literature sometimes applies finance-inspired metrics. A simple approach is to compute an *expected NPV* = $p_{\text{success}} \times (\text{net benefit}) - \text{cost}$. More sophisticated is to treat mission continuation as an option: e.g. if partial success allows step-2 investment (see real-options frameworks ⁷). Payne & Desai (2005) use decision trees with “risk-neutral” probabilities $q = (1-d)/(u-d)$ to weight outcomes ⁷. The result is an expected NPV view that discounts downside scenarios. In practice, very little public literature exists on deep-space ROI models. Most programmatic decisions in NASA use fixed budgets and look at cost per scientific merit (not strict NPV). However, some papers (e.g. “Risk-adjusted mission value,” Sadik et al. 2004) formulate a RAMV metric. Our open tool could include a risk-adjusted return calculator (for example, allowing users to specify p_{success} for each stage and computing an expected return).
- **Mission Insurance:** No major deep-space mission has publicly insured its payload beyond launch insurance. Industry insiders note that “many operators are choosing to launch without any insurance at all” due to high premiums ⁸. Launch insurance (if purchased) typically covers loss on ascent (often 5–20% of payload value), but after launch satellites/probes are rarely covered. NASA and JAXA missions do not budget insurance; losses (e.g. *Mars Observer*) are treated as program risks. ESA’s legal mandate forces operators to insure liabilities (not the spacecraft). The high market losses in 2023–2024 ($> \$500\text{M}$) have driven many commercial operators to self-insure ¹¹. The open literature contains scant data on deep-space mission insurance (no published premiums). The conclusion: mission planners should assume no insurance coverage for success – i.e. a failure means 100% loss of mission value.
- **Redundancy Strategies:** Missions typically build in redundancy for critical functions (e.g. cross-strap harnessing, duplicate computers) and consider redundant vehicles or launches. Well-known examples: the Parker Solar Probe had dual ~ 100 W solar arrays (failure tolerant); the Europa Clipper (planned) has dual strings for avionics and propulsion. At the spacecraft level, twin-satellite missions (e.g. GRACE, TandEM/ELENA proposals) send two identical craft in case one fails. Dual launches (rideshare of two probes) have been proposed to hedge an expensive payload. Analyses show that adding parallel modules increases reliability only as $P = 1 - (1-p)^n$. For instance, two identical thrusters in parallel yield $1 - (1-p_{\text{thruster}})^2$ success vs p_{thruster} . If $p_{\text{thruster}} = 0.99$ (99%), dual increases system to 99.9999%. However, the cost/mass roughly doubles. References on cost-reliability tradeoff (e.g. cost-benefit analyses of redundancy) exist in systems-engineering literature (e.g. Collopy’s and others’ NASA studies). In summary, the open-source planner should allow the user to specify redundancy levels (e.g. $N+1$ spares) and automatically compute resulting reliability-vs-cost. Our findings: current tools let you specify redundancies but rarely expose the resulting $P(\text{success})$ or cost impact directly to the user – that is a gap we could fill.

Multi-Mission Portfolio Optimization for Astrodynamics

- **Portfolio Optimization Literature:** The idea of treating mission planning as a portfolio (trade-offs among cost, schedule, risk, science return) has grown. Traditional references are few, but relevant work includes Conway's *Spacecraft Trajectory Optimization* (Cambridge Univ. Press, 2010) which covers individual mission optimization, and more recent "Architectures" research that uses multiobjective optimization. For example, O'Neill et al. (2020) describe a "Programmatic Portfolio Optimization" method for space architectures ¹². They model life-cycle phases and interdependencies of lunar exploration systems in a robust optimization framework. Mixed-Integer Programming (e.g. Gurobi, CPLEX) can encode logical constraints (if you send orbiter, you need lander; or limit on #launches). Metaheuristics (genetic algorithms, simulated annealing, particle swarm) are common because of the combinatorial nature of selecting sets of missions. Multi-objective evolutionary algorithms (NSGA-II, etc.) are used when trade-offs between multiple metrics (cost, science, risk) are needed. Many portfolio problems map to variants of the knapsack or TSP-like problems: e.g., "choose which asteroids to visit under a cost budget". Canonical references on multiobjective portfolio in space are still emerging, but recent Space Congress and AIAA papers cover examples (e.g. ERCs by Weber or Conway et al.).
- **GTOC Competitions:** The Global Trajectory Optimization Competition (GTOC) is a biennial "trajectories cup" that has galvanized multi-target mission methods ¹³. Each edition's problem (posted on the ESA GTOC portal) involves visiting multiple targets under tight constraints (e.g. GTOC5 – 15 NEAs with limited Δv , GTOC9 – multi-Moon explorers, GTOC12 – asteroid mining missions ¹³ ¹⁴). Winners share their methods in proceedings and reports. A consistent theme: teams use tree-search heuristics (beam search or branch-and-bound) seeded by greedy metrics, followed by continuous trajectory optimization. Bio-inspired and evolutionary heuristics (ant colony, genetic algorithms, simulated annealing) are also common for sequence design. For example, literature notes "beam search and its variants have been widely used with favorable results in GTOCs 4–9" ¹⁵. Ant colony (ACO) and GA are also "mainstream" for these time-dependent TSP problems ¹⁵. GTOC winner papers (many on arXiv or conference proceedings) detail these algorithms. **Key citations:** The ESA GTOC Portal ¹³ lists official problems; research abstracts ¹⁴ describe the widespread use of multiobjective and stochastic algorithms in these competitions. In particular, a recent arXiv note for GTOC12 ("Asteroid Mining" hosted by Tsinghua) shows modern beam/ant-colony hybrids. Learning from these, a mission-planner tool should incorporate multi-target tour optimization features, either via calling open solvers (e.g. NASA's OTIS, or pygmo with genetic heuristics).
- **Asteroid Tour Optimization:** Multi-rendezvous tour design is an active subfield. Classic work includes Siouris et al. (2004) on "Multiple Asteroid Tour", Gao's beam-search methods, and more recent "diversity-based strategy for asteroid tour design" (Bonnal et al. 2026, Sci. Direct) which notes that JPL's GTOC5 team used beam search to find an optimal asteroid tour ¹⁴. (Open-source tools like NASA's GMAT or ESA's pyKEP can be used to propagate low-thrust transfers between asteroids.) The GTOC problems themselves are basically multi-asteroid tour optimizations – their publications (e.g. Zhao et al. 2019, "JPL GTOC5 Results") show use of pseudospectral solvers and beam search ¹⁴. There are also papers on multi-asteroid tour under low-thrust constraints (e.g. Rollinson & Conway, 2017). For asteroid mining portfolios, GTOC12 (2023) is essentially such a problem; its solutions

(OptimiCS and others) used greedy heuristics then continuous optimization. **Portfolio constraints:** realistic mission portfolios must consider ground rules: e.g. at most 1–2 launches/year for big missions (since NASA/ESA budgets allow only ~1 flagship per few years); spacecraft manufacturing throughput (e.g. a large probe takes ~5–7 years to build from start); capital limits (e.g. NASA planetary budget ~\\$20–25B/decade total); diversity (mix Earth/Mars/comet missions). Decadal Surveys implicitly encode diversification. These constraints can be modeled as resource limits in the optimization (max #craft per year, cumulative cost cap). User surveys and org reports (e.g. NASA's 2011 Decadal Survey) note that, for instance, only one new Mars sample-return launch is feasible in a 12-year window. An open planner could allow the user to set these “cadence and budget” constraints and then solve for an optimal mix of missions.

Mission Planner UX and Tooling

- **Existing Tools:** There is a spectrum of mission-design software. **GMAT (NASA):** Open-source (Apache 2.0) tool with a GUI and script interface ¹⁶. It provides mission timeline, maneuver specification, multi-body gravity, and Monte Carlo. Its strengths: well-validated orbit mechanics, batch simulation, and orbit determination. Shortcomings: GUI can be unintuitive (especially for novices), limited built-in GUIs for custom optimization, and relatively heavy to install. **STK (AGI/Ansys Systems Tool Kit):** Commercial “gold standard” with extensive 3D visualization, timelines, sensor models, and scripting. Users praise its polished interface and rich physics libraries (coverage, RF link budgets, dynamic visualization), but note it is proprietary and expensive. **JPL Copernicus/MONTE:** Proprietary software stack (Copernicus at JSC; MONTE at JPL ¹⁷) used internally for high-fidelity trajectory design. Copernicus has a modern Python GUI and extensive interplanetary models (SRP, CR3BP). It was NASA's 2021 Software of the Year ¹⁸. MONTE is highly capable (supports Kalman filtering, batch orbit determination) but only available by license. Copernicus is free to US government/universities ¹⁷. **ESA Tools (GODOT, jeod, pygmo):** ESA's Goddard Space Flight Center's OSS repository includes “GODOT” (a design toolkit for satellite analysis) and uses ORBIT dynamics libraries; ESA also co-develops pygmo (a global optimization library used by Euclid's trajectory). **Open-source libraries:** *poliastro* (Python orbital mechanics), *PyKEP* (interplanetary solvers, used in GTOC9), *PyGMO* (multiobjective optimizers by ESA/ESAC). Also “orbital-mechanics-for-engineering-students” (likely referring to the Gilbert textbook code examples) and NASA's OpenMDAO for trades. These tools are scriptable (Python/C++) rather than GUI-driven. **Summary:** Tools differ widely in UX. Some conventions: almost all use vector/matrix inputs, a 3D view panel, ephemeris libraries, and timeline charts. Common workflows involve defining spacecraft and maneuvers, generating pork-chop plots or target lists, and iterating.
- **Visualization Patterns:** Standard plots abound in mission design. *Pork-chop plots* (Δv or C3 vs. launch date vs. TOF) are universal for launch/transfer planning. Δv vs. C3 vs. TOF *scatter plots* are often shown (e.g. clustering of solutions in GTOC publications). A mission planner should include interactive 2D plots (launch window contours) and scatter charts (e.g. Δv vs. trip time for many target choices). *Gantt/timeline charts* are widely used for scheduling phases (design, assembly, launch, cruise, operation). Tools like STK and GMAT offer timeline panels for maneuvers; open planners could emulate this (e.g. horizontal bar charts for mission phases). *3D trajectory visualization* (static or animated) is crucial: almost every tool displays the heliocentric or planet-centered path (COPERNICUS has built-in 3D preview ¹⁹). *Resource budgets:* mission designers often track fuel margin, power usage, data volume, etc. While no single “convention” exists, common dashboards

show time series of fuel and power (e.g. burn graphs, battery SoC). Incorporating these as real-time plots (with uncertainty shading) could improve UX.

- **Mission Plan Formats:** There is no single “mission-plan file format” widely shared. However, standards exist in related areas: NASA’s SPICE toolkit uses Spacecraft and Planet Kernel (SPK) files for trajectories and CK for attitudes, which are de facto exchange formats for orbit data. The CCSDS (Consultative Committee for Space Data Systems) issues standards like “Orbit Data Messages” (ODM) and “Trajectory Data Messages” (TDM) which define how to record ephemeris and maneuver sequences. For complex missions, some agencies use custom JSON/XML schemas (e.g. NASA’s internal “Cosmos” format or ESA’s Mission Planning CDF). In practice, planning deliverables are often PDF reports with diagrams and tables. An open planner should support exporting spacecraft orbits as SPK kernels or CCSDS OMM/TPK files, and perhaps import NASA-style mission configuration (e.g. spacecraft mass breakdown, timeline) via JSON.
- **Documentation Conventions:** A credible mission plan deliverable resembles a Phase A/B study report. NASA concept docs (e.g. Discovery or New Frontiers proposals) typically include sections: objectives, architecture, mission profile, system description, risk analysis, schedule, and cost. They use standard figure styles (e.g. labeled block diagrams, orbit plots, tables of key parameters). ESA’s Concurrent Design Facility (CDF) reports (when public) follow a template: summary of mission goals, timeline, system elements (mass/power budgets), and a generation of trade studies (often in annexes). Key elements include: pork-chop plots for launch windows, ΔV budgets table, timeline charts, and flight path diagrams. Finally, any open planner’s documentation should mimic these: provide exportable graphics and data tables that fit into a CDR/PD overview.
- **Adjacent UX Inspiration:** Financial portfolio tools and engineering tradespace interfaces offer useful patterns. For example, Bloomberg terminals use highly interactive dashboards with multiple synchronized views and keyboard shortcuts; while FactSet provides custom scatter-plot matrices and filters to sift thousands of securities. In engineering, NASA’s Trade Space Visualizer (ATSV) ²⁰ is a Java tool for high-dimensional data: it uses linked scatter plots, parallel coordinates, and “steering” (brushing) to explore large solution sets. Mission planners could borrow these ideas: e.g. allow the user to brush-select a cluster of candidate trajectories on one plot and immediately see its impact on cost and science metrics on another. Other useful patterns: auto-complete search for asteroids or planets (like Google’s quick find), live sliders for delta-v limits or launch dates, and small-multiples views comparing alternate mission scenarios side by side. Real-time feedback (e.g. updating a fuel budget as you adjust orbit parameters) can make planning more intuitive.

Tools to Benchmark: We should test against *GMAT* (NASA), *STK/Systems Tool Kit* (AGI), *Copernicus/MONTE* (JSC/JPL), and ESA’s *GODOT* libraries. Among open-source: *poliastro*, *PyKEP*, *PyGMO*, *OREKIT*, *Orbital Mechanics* (Gilbert), and the PSU *TradeSpace Visualizer* ²⁰. Each has strengths (GMAT’s validated propagation, STK’s UI, Copernicus’s solver, Poliastro’s ease-of-use) we should understand.

Key Papers and Codebases: Foundational references include Conway (2010) on trajectory optimization; NASA’s PRA handbook ¹; O’Neill’s portfolio PhD thesis ¹²; GTOC winning solution papers ¹⁴ ¹³; recent reviews on space mission risk and tradespace (e.g. Flypax’s 2024 “Space Mission Risk and Trends”). Important software to study: *GMAT* (GitHub), *Poliastro* and *PyKEP* (Python), *OpenOrbital* or similar, *NASA TReX*, and (for UX) *ATSV* and *Tableau-like* visualization frameworks.

UX Patterns to Emulate: Our planner should incorporate **interactive plots** (pork-chops, ΔV vs time matrices), **dashboard layouts** (multiple linked charts), **3D fly-through** of trajectories (WebGL or Unity), and intuitive **widgets** (date pickers, sliders for margins). Saving and sharing “profiles” (like Bloomberg watchlists) could allow mission teams to compare options. We should ensure the tool **makes uncertainty visible** (e.g. shading in timelines for possible delays, error bars on trajectory). Inspiration can be drawn from financial apps (clear numeric vs graphical emphasis) and engineering trade-tools (slider-driven “what-if” exploration).

Gaps in Current Tools: Most existing mission planners are either closed-source (STK, Copernicus) or focused on single-mission analysis (GMAT, Python libraries). Few offer *portfolio-level* optimization or integrated risk views. Uncertainty is typically treated as a bullet point, not surfaced to the user: e.g., GMAT doesn’t show probability of mission failure in its UI. Documentation is often locked in PDFs; there’s no unified import/export standard. We identified that **transparency** and **accessibility** are lacking: code and data behind trajectory solvers are not openly inspectable in proprietary tools. Also, the UX of these tools is geared toward experts (spreadsheets, scripts) rather than interdisciplinary teams. An open-source planner could fill this gap by providing both rigorous backend analysis (PRA, optimization algorithms) *and* an intuitive interface with honest visualization of confidence intervals and trade-offs.

Our Niche: We build an open, transparent mission planner that **embraces uncertainty** and prioritizes accessibility. We should deliver a competitive UX (inspired by Bloomberg-like dashboards and NASA tradespace viewers) with built-in probabilistic analysis (Monte Carlo, FTA) and multi-target planning capabilities (GTOC-style tour search). By benchmarking existing tools (GMAT/STK feature sets, Copernicus solvers, PyGMO algorithms) and adhering to mission-doc conventions (CCSDS/SPICE exports, NASA/ESA reporting styles), we ensure credibility. In summary, the state of the art offers many specialized components (robust propagators, UI widgets, optimizers) but no single open tool combines them with clear uncertainty communication. Our effort can fill that gap by uniting mission design, risk modeling, and interactive visualization in one open package.

Sources: Authoritative sources include NASA and ESA PRA guides ¹ ² ; historical mission reliability summaries ³ ⁴ ⁵ ; Copernicus and GMAT documentation ¹⁹ ¹⁶ ; GTOC publications ¹⁴ ¹³ ; and UX references such as the Trade Space Visualizer ²⁰ . These inform the above synthesis. (Any missing or unknown data is noted as gaps for future research.)

¹ ⁹ NPR 8705.5A Technical Probabilistic Risk Assessment (PRA) Procedures for Safety and Mission Success for NASA Programs and Projects (w/Change 2)

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